Emre Tiryaki

Backend Developer

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Professional Summary

Software engineer focused on backend development with Go and TypeScript. Experienced in building projects involving microservice architectures, distributed systems, and asynchronous messaging. Developed inter-service communication and real-time data processing infrastructures using technologies such as RabbitMQ, Redis, and GraphQL. Interested in scalable backend systems and maintainable software architectures.

Education

Inonu University Malatya, Türkiye
B.Sc. in Software Engineering *2023 – 2027*

Technical Skills

Programming Languages	Go, TypeScript, JavaScript
Backend	Gin, NestJS, Fastify, Express, RabbitMQ, Redis, GraphQL
Databases	PostgreSQL, MySQL, SQLite, MongoDB, Cloud Firestore
DevOps & Cloud	Docker, Linux, Shell Scripting, Git, AWS (EC2, Lambda, S3)

Experience

Avikon – Backend Developer Intern August – September 2026

- Independently designed and developed ClearSky, a real-time aircraft tracking system based on a microservice architecture.
- Developed a data ingestion service in Go to retrieve flight data from the OpenSky Network API and distribute it through RabbitMQ.
- Built an API layer using TypeScript, Fastify, and GraphQL to support real-time data querying and delivery to clients.
- Implemented the data access layer using MongoDB for persistent storage of selected aircraft positions, reporting, and flight route playback.

Competitions & Hackathons

Autonomous Finance Assistant (OFA) – Intelligent Payment Routing and Issue Resolution July 2026
Moka United AI Ideathon & Hackathon JavaScript, React, Express, Python, Machine Learning

- Developed a machine learning-based optimization algorithm to identify the most suitable acquiring bank among multiple alternatives.
- Designed an AI agent architecture capable of detecting system errors, autonomously generating solutions, and taking corrective actions.

OKUR – AI-Powered Exam Scanning and Assessment System May 2026
Inufest 2026 Competition TypeScript, NestJS, Python, Machine Learning, React, Redis

- Developed a system that automatically generates exam questions in LaTeX format.
- Built a module that processes exam papers using computer vision techniques and converts handwritten student answers into text.
- Integrated an AI model that automatically evaluates student answers by semantically comparing them with instructor-provided answer keys.

Cosmic Data Processing Pipeline March 2026
TUA Astro Hackathon Python, FastAPI, React, NumPy, SciPy

- Developed a component that generates synthetic cosmic ray data in real time and feeds it into the processing system.
- Built a real-time digital signal processing (DSP) pipeline using NumPy and SciPy vectorization techniques to mitigate signal distortions caused by cosmic radiation.

HEGS News – Gamified News Platform October 2025
Anadolu Agency Media Technologies Hackathon JavaScript, Java, Spring Boot, Python, FastAPI

- Developed the frontend of a platform that presents Anadolu Agency news through a user-friendly interface and automatically generates AI-powered summaries.
- Implemented user profiles, friend management, and mini-games as part of the gamified news experience.

Open-Source Contributions

- ani-cli #1938 – Exact Match Flag: Added `-exact` and `-exact-match` options to the open-source anime search and streaming tool, enabling case-insensitive exact title matching in search results. The contribution was reviewed, accepted, and merged into the upstream repository.
- ani-cli #1975 – Temporary Download Directory: Implemented a feature that separates temporary files generated by `yt-dlp` from the target download directory and allows the temporary directory to be configured through an environment variable. Pull request is under review.

Independent Projects

- RepoGrab: Built a Go-based TUI tool that allows downloading individual files from GitHub and GitLab repositories without cloning the entire repository.
- auto-readme-generator: Developed a tool that analyzes project structures and generates README files using user-selected AI models. Built the backend with Rust and Axum, and the frontend with React and JavaScript.
- pdf2markdown: Developed a Python CLI tool that converts PDF files into Markdown through parallel processing using locally hosted models via Ollama.
- project-toolkit: Developed a Bash-based command-line tool that automates project creation and directory organization workflows.

Diller

Turkish Main Language

İngilizce C1 – Advanced